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Final Project Paper: Post-COVID Math Achievement Gaps

Introduction

This report contains the modeling strategy, results, and implications of an empirical study intended to examine the post-COVID implications on student outcomes and gaps. The study has data from 100,000s of districts across the United States. The data is from the Stanford Education Data Archive (SEDA) and is representative of grades 3 through 8.

> **Research Question 1:** How does math achievement levels post COVID-19 differ across race, gender, and grade level?
>
> **Research Question 2:** To what extent are post COVID-19 math achievement gaps by race and gender affected by socioeconomic status after controlling for grade level and year?

All achievement scores are represented in both standard deviations and years behind based on Hanushek's (2019) research on racial achievement gaps. A standard deviation of zero represents the national average, negative SDs represent a below average math achievement score, and positive SDs represent an above average math achievement score. All tables will show both a SD and a 'years behind' value to represent how long it would theoretically take for the students in that group to catch up to the national average of 0 standard deviations. That data was split into 2 categories which include pre-COVID (2017-2019) and post-COVID (2022-2024).

Literature Review:

The COVID 19 pandemic caused one of the most pervasive educational disruptions to the United States in modern history. In the Spring of 2020, millions of K-12 students abruptly switched from face-to-face instruction to remote instruction, many of which remained remote or partially remote for the 2021-2022 school year (Levinson & Markovits, 2022). Data from 7,800 school districts found that losses were far more profound in low income and minority students and test scores were lower for districts whose students spent more time in 2020-2021 in remote learning (Fahle et al., 2025). Additionally, "Large scale evidence from the U.S. consistently shows that the pandemic disproportionately impacted the reading and math achievement of students who are classified as Black, Hispanic, American Indian or Alaskan Native and those from high poverty schools" (Gee et al., 2023; Golden et al., 2023). According to Frances & Weller (2022), Black students were, "less likely to have reliable access to remote devices, which decreases remote study time." Another emerging theme since COVID is that student achievement scores are also affected by what grade the student was in during the 2019-2020 school year. This has a huge impact on whether their achievement scores increased or decreased along with race and socioeconomic status. Students in younger grades tended to regain some of their learning loss

while students in older grades showed fewer gains, more specifically, “the third through fifth grade cohorts regained between one-quarter to one-third of their 2020–21 loss, while the sixth grade cohort regained only about 16 percent” (Bauer & Powell, 2025). For all the reasons stated above, there should be a focus on why grade, race, and gender play such huge factors in math achievement, what those gaps were caused by, and how those achievement gaps can be closed.

Data Preparation

The full dataset includes over 1.7 million observations from years 2009 to 2024. Several steps were taken to properly prepare the data for analysis.

Filtering: The analysis was restricted to math achievement from grades 3-8 yielding 852,141 observations. Other subjects such as reading or literature scores were excluded. Observations were then labeled by time which included pre-COVID (2017-2019, n=185,558) and post-COVID (2022-2024, n=162,591), any years outside of 2017-2019 and 2022-2024 were filtered out.

Achievement Gap Identification: Three achievement gap variables were constructed by subtracting the mean score of the reference group from the focal group at the district, grade, and year level. Negative values indicate focal group scores lower than the reference group. The reference groups are White and Male student groups because they are the higher performing groups and the focus groups are Black, Hispanic, and Female student groups which are shown in the analysis below. The gaps constructed were: Black minus White, Hispanic minus White, and Female minus Male. Gap observations were retained only when both subgroups had non-missing scores. The resulting sample sizes ranged from 26,207 observations for the Black-White gap to 93,471 for the Female-Male gap, reflecting a significant distribution of gender disaggregated reporting to race-disaggregated reporting.

Socioeconomic Status Measure: The primary indicator used in all regression models is the SEDA ‘sesavgall’ variable which measures household income, parental education, poverty rates, unemployment, and single parent household rates into an index. The variable had the lowest missingness of all the socioeconomic measures available at 4.7% making it the most useful and accurate variable for socioeconomic status (SES).

Overall Achievement Gaps

Research question 1 asks how math achievement post-COVID differs across race, gender, and grade level. The data shows where gaps exist, how large those gaps are, and whether there are any changes post-COVID, while although not a model, it does provide valuable information from the dataset. Table 1 represents mean math achievement scores by student subgroups during the post-COVD period. All values are in SD units and years to recover based on Hanushek’s research (2019). Hanushek et. al. state that “1 standard deviation is approximately the difference in the average performance of students in 4th and 8th grades, or four years’ worth of learning”

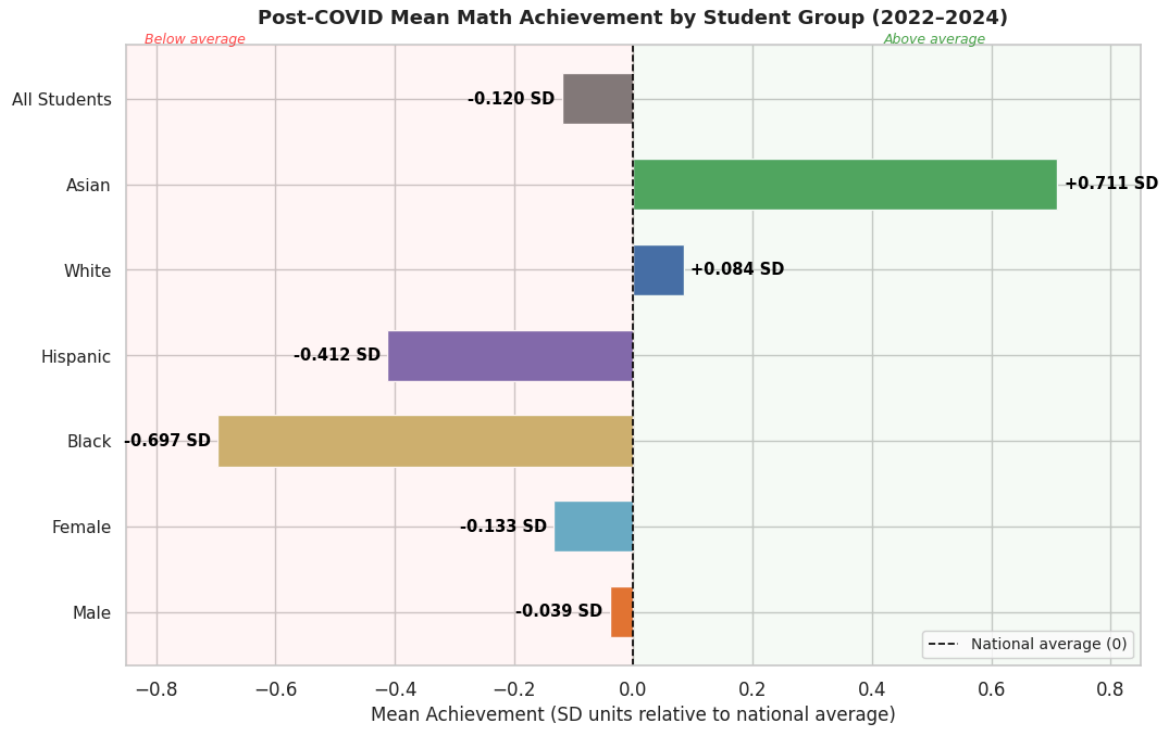
(2019). For the purposes of this paper, 1 standard deviation will represent 3 years. Although not the racial group focused on in this paper, Asian students performed the highest. Despite this fact, it helps to compare the Asian student group with all other racial groups to show the overall gaps between groups. Black students performed lowest at about 2 years behind White students. The overall average of -0.120 SD indicates that post-COVD math achievement nationally remains below pre-pandemic benchmarks. This average comes from SEDA's weighted composite student score which is an average from the years 2009 to 2024. The table below shows that all students are at an average of -0.120 SDs meaning they are 0.120 SDs below the average from the composite average from 2009-2024.

Table 1

Student Group	Mean Achievement in SDs	Years to Average	Number of Districts	Interpretation in relation to national average
All students	**-0.120**	-0.36	162,591	Below
White	**+0.084**	+0.25	122,066	Above
Black	**-0.697**	-2.09	34,101	Largest deficit
Hispanic	**-0.412**	-1.24	55,241	Moderate deficit
Asian	**+0.711**	+2.13	17,742	High performing
Female	**-0.133**	-0.39	104,123	Small deficit
Male	**-0.039**	-0.12	105,493	Close to average

Note: Achievement is expressed in standard deviation units relative to national average (represented as zero). Years to average represents the number of years it would take this group to catch up (positive values represent years ahead and negative values represent years behind). Interpretation represents where each group falls in relation to the average (SD of zero).

Graph 1: Post-COVID Mean Math Achievement by Group



Graph 1 shows mean math achievement trends by race (2022–2024) and by gender (2022–2024). The break in the y-axis is represented by the national average, a standard deviation of 0.

Grade Level Achievement Gaps

Table 2 shows mean achievement gaps by grade during the post-COVID period. The trends that stand out are the Black-White and Hispanic-White gaps. Both are large and stable across grade levels with standard deviations between -0.68 and -0.74 standard deviations for the Black-White gap and between -0.47 and -0.54 standard deviations for the Hispanic-White gap. Both gaps show modest growth by 8th grade, though the disparities are still substantial. The Female-Male gap is also supported by Kuhfeld's research as enrollment for girls in algebra decreased by 2.9% compared to a 1.4% decline for boys (2025). It's important for district leaders to understand why girls are not enrolling in higher level math classes to attempt to close the gender gap that has widened significantly post-COVID. Additionally, district leaders can focus on encouraging girls to take high level math classes, we know it is a math specific issue because girls' reading scores were affected far less than math scores (2025).

Table 2: Achievement Gaps and Years to Recover

Grade	Black-White Gap	Years to Average	Hispanic-White Gap	Years to Average	Female-Male Gap	Years to Average

3	-0.737	-2.21	-0.537	-1.61	-0.133	-0.39
4	-0.744	-2.23	-0.524	-1.57	-0.137	-0.41
5	-0.707	-2.12	-0.487	-1.46	-0.080	-0.24
6	-0.713	-2.14	-0.497	-1.49	-0.039	-0.12
7	-0.732	-2.20	-0.511	-1.53	-0.046	-0.14
8	-0.676	-2.03	-0.471	-1.41	+0.052	+0.16

*Note: all gap values are in standard deviation (SD) units. Negative values indicate scores lower than the average. Years to average lists the number of years it would take for the student group to "catch up" to the national average. A negative year to average represents how many years behind. Years behind is calculated by (gap*3 years)*

Pre-COVID vs Post-COVID Comparison

Table 3 shows mean achievement gaps from pre-COVID and post-COVD. The gaps worsened in post-COVID for every category. The Black-White gap widened by 0.055 standard deviations and the Hispanic-White gap widened by 0.036 standard deviations, which suggests that Black and Hispanic students experienced greater setbacks relative to White students. The most striking finding is related to the gender gap. Prior to COVID, female students were performing at about the same level as male students (+0.007 standard deviations), a difference of nearly 0. According to Kuhfeld et. al., "by 2019, data from an international math and science assessment showed that the historic advantage eight-grade boys held in math and science achievement had largely disappeared" (2025). Post-COVID, female students fell 0.095 standard deviations below male students, a reversal of 0.102 standard deviations. This shift warrants a particular attention from school and district leaders to develop recovery strategies to help close the math gap for female students.

Table 3: Achievement Gaps Comparing Pre and Post COVID Differences

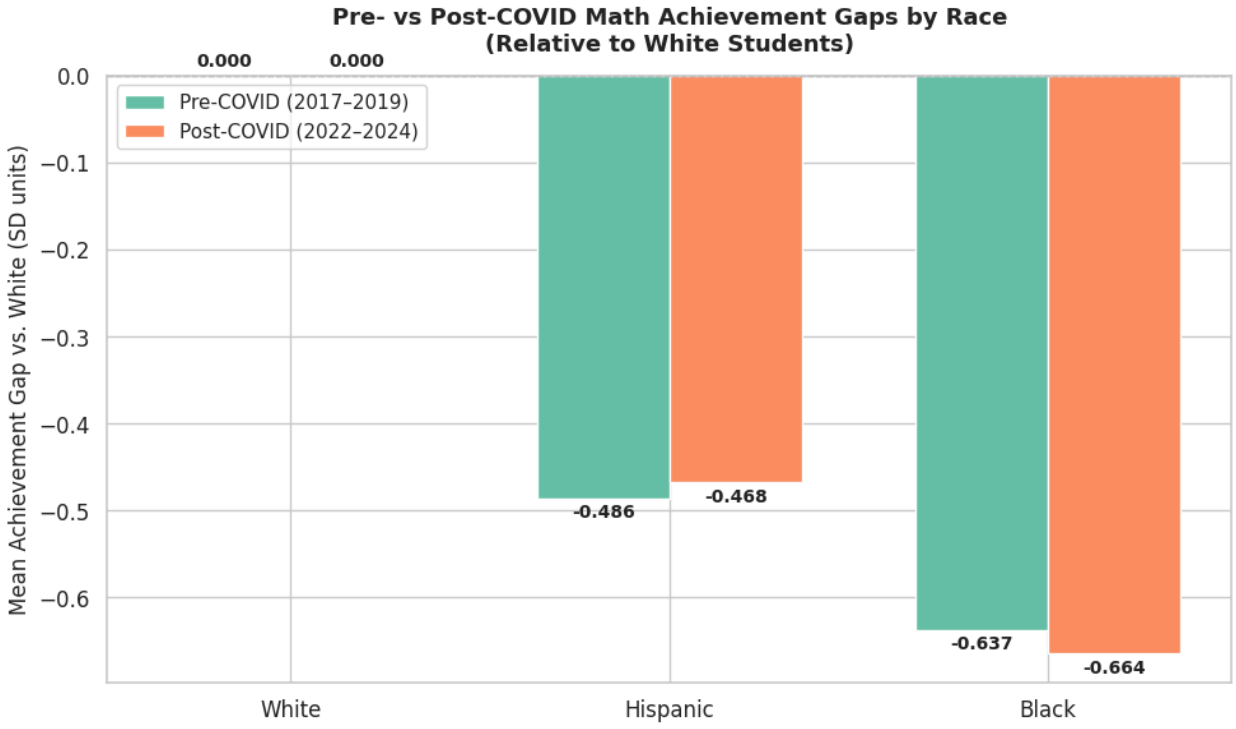
Gap	Pre-COVID (2017-2019)	Post-COVID (2022-2024)	Difference between pre/post COVID	Difference in Years
Black-White	-0.667	-0.722	-0.055	-0.17 years
Hispanic-White	-0.447	-0.482	-0.036	-0.11 years
Female-Male	+0.007	-0.095	-0.102	-0.31 years

Note: Pre-COVID period (2017-2019); Post-COVID (2022-2024). Difference = Post-COVID minus Pre-COVID. Negative differences indicate a decrease from pre to post-COVID. Negatives years indicate how many years behind.

The Black and White student group comparison shows that the difference between pre and post COVID gaps widened by 0.17 years, in other words, this would represent 31 school days. The Hispanic-White and Female-Male groups widened by 20 and 56 school days respectively. All day values are calculated by using an average school year of 180 days. By comparison, most

summer schools are 4 to 8 weeks, or 20 to 40 full days. This is equivalent to 10-20 school days if each day is a half day and the student spends all 10 to 20 half days on a single subject. In most cases, if those students attempted to recover over the summer, there aren't enough days available between the regular school year and the summer to make up for between 20 and 56 school days. However, in the interest of recovery, it may be worth considering doing 4 to 8 weeks of full days (or 20 to 40 days) to help students recover. Additionally, high dosage tutoring at 5 hours a week throughout a 180 day school year (5 hours a week for 36 weeks) could provide students with about 25 days of recovery time. Implementing both summer school and high dosage tutoring throughout the school year could help students regain up to 65 school days in a single school year.

Graph 2: Pre vs Post COVID Math Achievement Gaps by Race



Note: zero represents the average

Focus for District Leaders

A gap of about 0.70 standard deviations post-COVID between Black and White students represents a profound disparity in learning outcomes that compounds over time. 1 standard deviation of achievement corresponds to about 3-4 years of learning. At best, Black students are about 2 years behind their White peers, a gap that has widened since COVID. Teachers,

administrators, and school districts must consider how to help students regain those 2 years within school hours. For Hispanic students, the outlook is slightly better with a gap of around 0.45 standard deviations or about 1.35 years. District leaders should consider preventative measures that are subgroup specific rather than overall interventions that are intended to fit all students.

RESEARCH QUESTION 2: THE ROLE OF SOCIOECONOMIC STATUS

Modeling Strategy

Research Question 2 was addressed using two complementary regression models. One looked at post-COVID and one compared pre and post COVID. Both use Ordinary Least Squares (OLS) regression with district clustered standard errors to account for districts having repeated observations for grades and years. Clustering grouped each district so that each observation from the same district is treated as a group. The unit of analysis used are the district, grade, and year observation. The outcome in each model is an achievement gap score, Black minus White, Hispanic minus White, or Female minus Male rather than individual student scores to measure the gaps between groups. The focus is how much higher or lower one group scored compared to another group. This is appropriate because SEDA does not provide student data and instead provides district level data. Conclusions therefore describe systemic patterns across districts.

Model 1: Post-COVID Regression

After controlling for grade level and year, the model estimates the connection between district socioeconomic status and achievement gaps during the post-COVID period (2022-2024). Separate models were created for each of the 3 sub comparison groups. Controlling for grade and year using fixed effects, each achievement gap was regressed on socioeconomic status so that the SES coefficient reflects a within grade and year relationship between district wealth and gap size.

Table 4. Post-COVID Gap Regression Results: SES and Achievement Gaps (2022–2024)

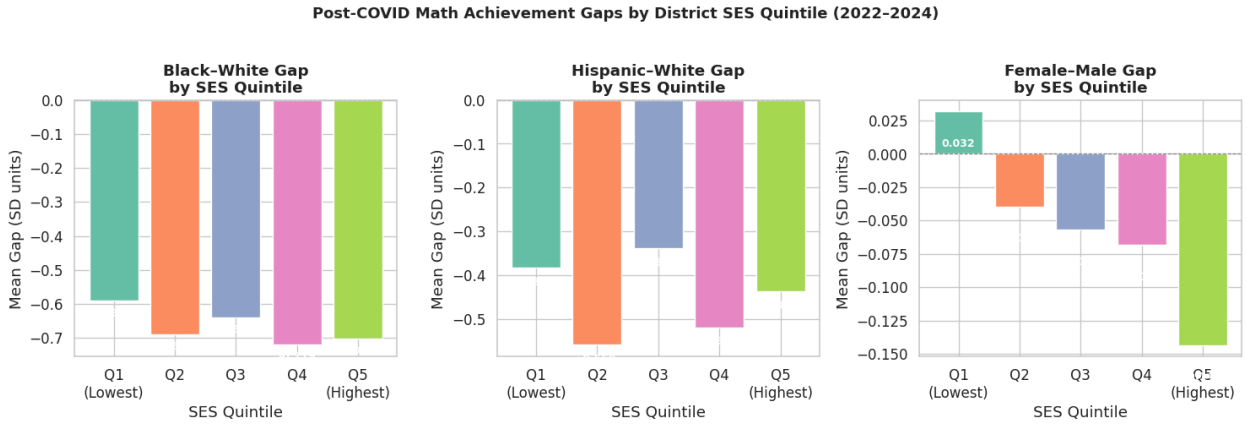
Gap Subgroups	SES Coefficient	p-value	N
Black–White	-0.0677	< .001	26,207
Hispanic–White	-0.0811	< .001	43,666
Female–Male	-0.0283	< .001	93,471

All three models produced statistically significant coefficients ($p < .001$). Districts with a higher SES are all associated with higher achievement gaps. For Black-White gaps, each 1 unit increase in district SES is associated with another 0.068 standard deviation (or 37 school days behind their White peers). For the Hispanic-White group, the gap widened by 0.081 standard deviations, the largest of the 3 groups, representing 44 school days behind their White peers. Lastly, the

Female-Male gap represents 0.028 standard deviations or 15 school days behind their White peers. Despite the Female to Male gap having the smallest standard deviation, 0.028 is still highly significant.

In high SES districts, achievement is generally high. However, white students disproportionately benefit from these districts. Black and Hispanic students with similar socioeconomic status do not benefit or appear to have equal access to the same advantages as their White peers. Additionally, Male students disproportionately benefit from being in these districts compared to Female students.

Graph 3: Post-COVID Math Achievement Gaps by SES Quintile



Note: The quintiles represent districts and are split into 5 groups from lowest 20% to highest 20%. The graph shows that for Black students and Female students, the gap increases as SES increases.

Model 2: Pre to Post COVID Model

The second model was chosen to determine whether districts with different SES levels experienced significant changes pre and post COVID. District grade averages were calculated for pre and post COVID periods, the change was noted for each gap, and the model was a function of COVID SES and grade fixed effects:

$$\Delta \text{ Gap} = \beta 0 + \beta 1(\text{Pre-COVID SES}) + \text{Grade Fixed Effects} + \varepsilon$$

The formula asks whether gaps improved or worsened after COVID. A positive coefficient represents a higher SES district with a greater widening after COVID and a negative coefficient indicates that a higher SES district saw a narrowing.

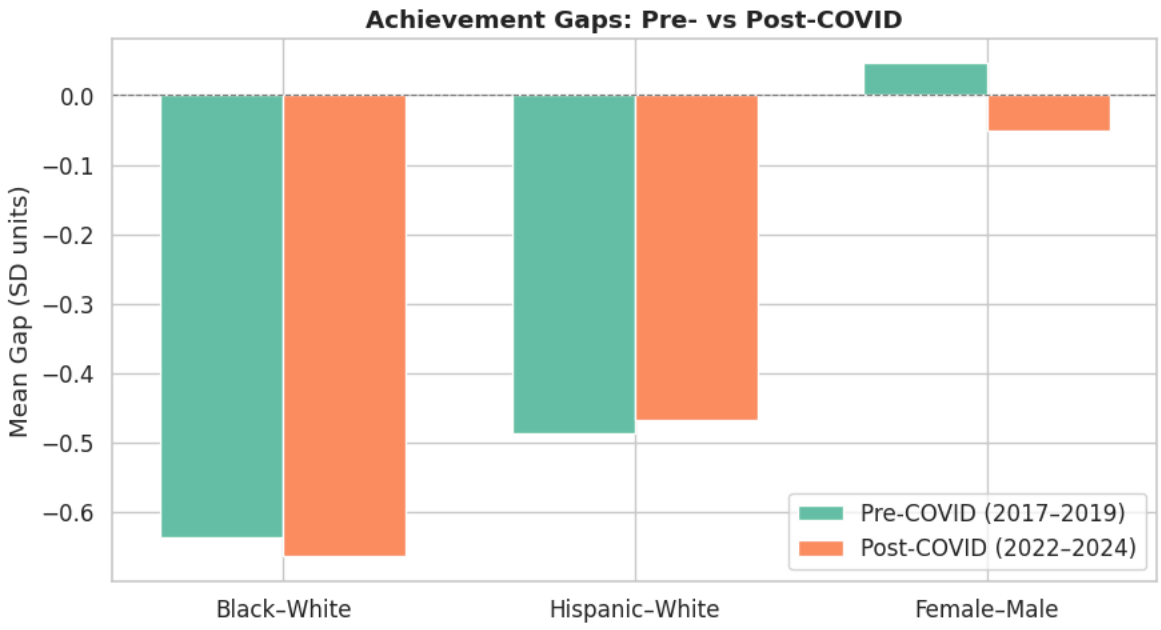
Table 5. Pre-to-Post COVID Change Model Results: SES and Gap Changes

Gap Subgroups	SES Coefficient	p-value	N
Black–White	-0.0092	.010	9,458
Hispanic–White	+0.0051	*.103*	15,227
Female–Male	+0.0075	< .001	34,806

Note. Outcome is post-COVID gap minus pre-COVID gap at district-grade level.

Districts with a higher SES saw a widening of math achievement between Black and White students on average compared to lower SES districts at 0.009 standard deviations demonstrating that COVID resources helped White students recover faster than Black students. It is small (0.009 SDs) but worth noting that higher SES districts are also seeing the effects in their Black student population while their White population is either at or close to pre-COVID averages. White students overall have recovered faster from COVID school disruptions through other factors that could be attributed to things like access tutors and resources. This finding is supported by Assari & Zare who state, "positive effects of high family SES on school characteristics were significantly weaker for Black adolescents than for White adolescents" (2025). The Hispanic-White gap is not statistically significant at a p-value of 0.103. The gap widened similarly across all districts no matter the SES of those districts. The Female-Male shows a positive SES coefficient, meaning that although Female students fell behind Male students post-COVID, this did not happen in all districts equally or uniformly as it did for Hispanic students. In higher SES districts, Female students still fell behind, however, less than in lower SES districts. Wealthier districts did have some buffers for Female students to begin to recover from COVID faster than lower SES districts.

Graph 4: Pre vs Post COVID Achievement Gaps



Note: zero represents the average

Conclusion & Implications for Stakeholders

There are three conclusions that stand out among the results. First, racial achievement gaps in math are quite large with Black students performing the worst across all SES groups, grades, and years. Black students are at about 0.70 standard deviations below their White peers on average which represents about 2 years of recovery. District leaders should treat this disparity as a systemic emergency. Next, the gender gap in math reversed post-COVID. Prior to COVID, Female students who were on par with their Male counterparts, are now behind post COVID. This finding is most severe in lower SES districts. Recovery strategies should explicitly focus on addressing Female students' trajectory in math achievement, including entry into Honors and AP math courses. Lastly, higher SES districts are associated with wider racial achievement gaps. There is an assumption that sending Black and Hispanic students to higher income districts automatically means that resources are going to reduce achievement gaps. However, this is not the case. The data suggests that these resources are unequal across student groups. Focusing on equity and instructional strategies that benefit Black and Hispanic students in higher SES districts could decrease the math achievement gaps in those districts. Ensuring that higher SES districts have recovery interventions alongside other resources will allow a more equitable environment for all students.

These findings are relevant to district leaders, state education agencies, and education policymakers who are designing post-COVID recovery programs. Subgroup specific programs targeting Black, Hispanic, and Female students across both high & low SES districts, will help

close math achievement gaps. District leaders should avoid designing programs intended to work for all students, the results show that recovery strategies should be equitable and focus on subgroups who are the least likely to remain on track with the national average. Additionally, policymakers should focus on preventative measures to focus on maintaining a closed or minimal gap from elementary through high school. Further research should be done on what aspects of prevention can close gaps with an equitable recovery focus.

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